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# Fourier ptychography via wavefront modulation with a diffuser

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ABSTRACT

Aperture-scanning Fourier ptychography is a configuration of Fourier ptychgraphy by scanning an aperture at the back focal plane of the objective lens. It has been a useful tool for measuring the complex scattering field exited from a sample. However, in this method the intensity of dark images at the charge coupled device (CCD) plane is of same order or even lower than the noise when the high frequency of the Fourier spectrum is filtered by the aperture. Therefore, the recorded dark field images suffer from low signal-to-noise ratio (SNR), which results in poor reconstruction. In this paper, we report a modified Fourier ptychographic imaging method by wavefront modulation. Instead of an aperture, a diffuser is used to modulate the spectrum in the presented method. The complex distribution of the sample and the diffuser are reconstructed simultaneously via iterative phase retrieval. Numerical simulations and proof-of-concept experiments have been carried out to demonstrate the feasibility of the presented method.

## 1. Introduction

Fourier ptychography [1–13] is a newly developed powerful computational imaging framework based on the principles of synthetic aperture imaging [14–16] and iterative phase retrieval [17–21]. Since its first proposal, Fourier ptychography has been successfully demonstrated in the field of high-resolution imaging [22–24], quantitative phase imaging [25,26] and biomedical applications [27,28]. In the early implement of Fourier ptychography, a light emitting diode (LED) array is used to generate beams of varying angles [1] to illuminate the sample to be observed, and a CCD camera is placed at the image plane of the imaging system to capture the low resolution images which is filtered by the numerical aperture (NA). By assuming that the spectrum of the sample shifts in the Fourier space when the sample is illuminated by a titled illumination beam, the complex amplitude of the spectrum can be reconstructed rapidly and accurately through a common Fourier ptychographic algorithm relying on overlapping constraints and intensity constraints, hence the high resolution amplitude as well as the quantitative phase image of the sample can be obtained by inverse Fourier transform the reconstructed spectrum. However, the assumption is established under conditions where the sample is sufficiently thin, for thick sample the complex distribution of the spectrum changes when the sample is illuminated by a beam with different angle, thus the overlapping constraints are not satisfied and the common ptychographic algorithms will not be able to reconstruct the complex spectral. To address the thick-limitation problem in Fourier ptychography, Fourier ptychographic tomography (FPT) [29] and multiple-slice Fourier ptychography [30] inspired by three dimensional ptychography [31–33] were proposed for handling thick samples at the cost of dramatically increased computational complexity.

Aperture-scanning Fourier ptychography [34–37] is another method to bypass the limitation of sample thickness in Fourier ptychography. Instead of illuminating sample with varying angle beam, in the configuration of aperture-scanning Fourier ptychography, the thick sample is illuminated by a collimated plane wave. An aperture is placed at the confocal plane of the 4f imaging system to modulate the spectrum of the thick sample and multiple intensity images including the bright field images and dark field images are recorded by a CCD (or CMOS) camera when the aperture is shifting in the confocal plane. After data acquisition is completed, the recorded images are converted to the complex distribution of the spectrum by a conventional Fourier ptychographic algorithm [38,39] to recover the wavefront exited from the thick sample. Due to its convenient setup and low computational complexity, aperture-scanning Fourier ptychography has been used for phase imaging [36], topography [40], 3D refocusing, and pixel super-resolution [34,41]. However, the intensity of dark image at the CCD plane is of same order or even lower than the noise when the aperture is scanned to the high frequency position of the spectrum, hence the recorded dark field images suffer from low signal-to-noise ratio (SNR), resulting in poor reconstruction of aperture-scanning Fourier ptychography.

Recently, a diffuser was employed in imaging methods based on phase retrieval. Zhang et al. [21] proposed a phase retrieval method by

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coherent modulation imaging, in which a diffuser is inserted between the object and detector to modulate the wavefield. The method has proven to have the advantages of compact setup, fast data acquisition, and high convergence speed. However, in their method, the complex transmission function of the diffuser should be known accurately in advance. Xie et al. [42] proposed an improved scanning-aperture Fourier ptychographic method by placing a known phase modulator in the optical setup. The method was demonstrated with better convergence and robust to shift errors and noise by computer simulations, however the method still needs to scan the spectrum in the confocal plane using an aperture, the problem of low SNR of the dark field images were not avoided. In addition, it is difficult to accurately obtain the complex distribution of the phase modulator in the experiment. Zheng et al. [43,44] proposed pattern-illuminated Fourier ptychography and near field Fourier ptychography by illuminating a sample with shifted diffuser. However, their method cannot be used for thick sample unless multiple-slice algorithm [30] is combined.

In this paper, we propose a robust Fourier ptychographic imaging method via wavefront modulation with a diffuser. Instead of using an aperture, a diffuser placed near the con-focal plane is fixed on a translation stage and shifted laterally to modulate the complex spectrum of the sample in the proposed method. In the reconstruction, the amplitude as well as the phase of the complex sample and the diffuser are iteratively reconstructed simultaneously via a Fourier ptychographic algorithm. The manuscript is organized as follows: Section 2 describes the proposed method in detail; Section 3 shows the simulation results and Section 4 shows the experimental results to demonstrate the feasibility of the proposed method; and Section 5 concludes.

## 2. Method details

The original schematic setup of conventional aperture-scanning Fourier ptychography is illustrated in Fig. 1(a), which is built on the basis of a conventional 4f imaging system. In the configuration the sample is illuminated by a plane wave and a CCD detector is placed at the conjugate plane of the sample plane. An aperture fixed on a two-dimensional translation stage is shifted at the confocal plane of the 4f imaging system to filter different part of the sample spectrum. The intensity distribution formed at the plane of the detector is expressed as:

$$I_j(x,y) = \left|\mathcal{F}\left\{O(u,v)\,A_j\left(u-u_j, v-v_j\right)\right\}\right|^2, \tag{1}$$

where $(x, y)$ and $(u, v)$ are coordinates in the detector and con-focal plane, respectively. $\mathcal{F}$ is the 2D fourier transform operator. $O$ is the spectrum distribution of the wavefront exiting the sample, $A$ is the aperture function and $(u_j, v_j)$ represents the central position of the scanning step $j$. Different from the conventional method, our proposed method eliminate the usage of the aperture, as shown in Fig. 1(b), a diffuser is insert to modulate the spectrum. The intensity pattern recorded by the detector can be calculated as below:

$$I_j(x,y) = \left|\mathcal{F}\left\{\mathcal{P}\left[O(u,v), z\right] M\left(\xi-\xi_j, \eta-\eta_j\right)\right\}\right|^2, \tag{2}$$

where $M$ is the complex transmission function of the diffuser, $z$ is the distance between the diffuser and the con-focal plane. The phase modulator can be placed at any position near the confocal plane [41]. $\mathcal{P}$ represents the Fresnel propagation, which is defined as:

$$\mathcal{P}\{f(x,y), z\} = \frac{\exp[jkz]}{j\lambda z}\iint f(x,y)\exp \times\left[jk\frac{\left(x_0-x\right)^2+\left(y_0-y\right)^2}{2z}\right]\mathrm{d}x\mathrm{d}y. \tag{3}$$

After finishing the data acquisition, the reconstruction process is started with initial guesses of the exiting wavefront function $o(x, y)$ and the diffuser function $M$. The guess of $o(x, y)$ can be randomly distributed or obtained by summing all the collected intensity patterns while the initial guess of diffuser function can be randomly distributed. The $k$th reconstruction process is proceed as follows:

(1) Calculate the wavefront spectrum $O_k(u, v)$ by Fourier transform the guessed wavefront function:

$$O_k(u,v) = \mathcal{F}\left\{o_k(x,y)\right\}. \tag{4}$$

(2) Propagate $O_k(u, v)$ to the plane of diffuser:

$$T_k(\xi,\eta) = \mathcal{P}\left[O_k(u,v), z\right]. \tag{5}$$

(3) Calculate the wavefront $\psi_{j,k}(x, y)$ formed at the plane of detector:

$$\psi_{j,k}(x,y) = \mathcal{F}\left\{T_k(\xi,\eta)\,M_k\left(\xi-\xi_j, \eta-\eta_j\right)\right\}. \tag{6}$$

(4) Replace the amplitude of wavefront $\psi_{j,k}(x, y)$ using the square root of the recorded intensity $I_j(x, y)$ while keep the phase unchanged, an updated version of the wavefront $\psi_{j,k}(x, y)$ is obtained by:

$$\psi_{j,k}(x,y) = \sqrt{I_j(x,y)}\exp\left(i\theta_{j,k}\right), \tag{7}$$

where $\theta_{j,\,k}$ is the phase of $\psi_{j,k}(x, y)$.

(5) Inverse Fourier transform of $\psi_{j,k}(x, y)$ and an updated version of the wavefront existing the diffuser is:

$$\phi_{j,k}(\xi,\eta) = \mathcal{F}^{-}\left\{\psi_{j,k}(x,y)\right\}. \tag{8}$$

(6) Calculate the wavefront difference at the front plane of the diffuser:

$$\Delta_{j,k} = \phi_{j,k}(\xi,\eta) - T_k(\xi,\eta)\,M_k\left(\xi-\xi_j, \eta-\eta_j\right). \tag{9}$$

(7) Update the diffuser and the wavefront immediately before the diffuser:

$$\begin{aligned} M_{k+1}\left(\xi-\xi_j,\eta-\eta_j\right) &= M_k\left(\xi-\xi_j,\eta-\eta_j\right) + \frac{\left|T_k(\xi,\eta)\right|}{\left|T_{\mathrm{k,max}}(\xi,\eta)\right|}\frac{T_k^*(\xi,\eta)\,\Delta_{j,k}}{\left(\left|T_k(\xi,\eta)\right|^2+\alpha\right)} \\ T_{k+1}(\xi,\eta) &= T_k(\xi,\eta) + \frac{\left|M_k\left(\xi-\xi_j,\eta-\eta_j\right)\right|}{\left|M_{\mathrm{k,max}}\left(\xi-\xi_j,\eta-\eta_j\right)\right|} \cdot \frac{M_k^*\left(\xi-\xi_j,\eta-\eta_j\right)\Delta_{j,k}}{\left(\left|M_k\left(\xi-\xi_j,\eta-\eta_j\right)\right|^2+\beta\right)}, \end{aligned} \tag{10}$$

where $\alpha$ and $\beta$ are constant to control the update step size.

(8) Back propagate $T_{k+1}(\xi, \eta)$ to the focal plane and generate the complex wavefront exiting the sample by inverse Fourier transform:

$$o_{k+1}(x,y) = \mathcal{F}^{-}\left\{\mathcal{P}\left[T_{k+1}(\xi,\eta), -z\right]\right\} \tag{11}$$

(9) Repeat steps (1)-(8) until the sum square error ($SSE$) is sufficient small, the SSE is evaluated at the detector plane as:

$$SSE = \frac{\sum\left(I_j(x,y) - \left|\psi_{j,k}(x,y)\right|^2\right)^2}{N} \tag{12}$$

where $N$ is the number of pixels.

## 3. Numerical verification

In this section, we verify the proposed method by conducting computer simulations. Firstly, we performed ten sets of simulations by using different simulated samples. For each simulation, the image (consists of 512 × 512 pixels with a pixel size of 7.4 μm$^2$, the pixel values were scaled to [0–1]) in the 1st row of Fig. 2 was selected as the amplitude of the sample, and the image (consists of 512 × 512 pixels with a pixel size of 7.4 μm$^2$, the pixel values were scaled to [0–$\pi$]) in the 2nd row of Fig. 2 was selected as the phase of the sample. The parameters used in the simulations were selected to model a real

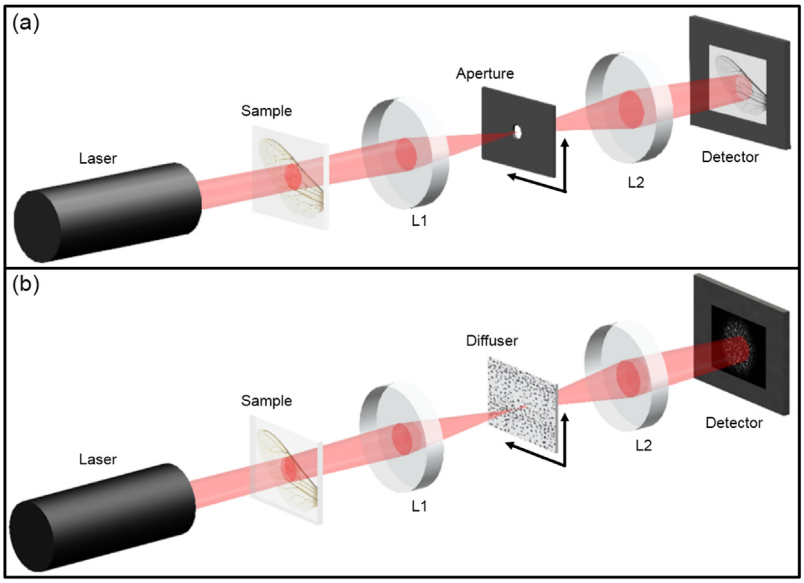


**Fig. 1.** Schematic setup of (a) conventional aperture-scanning Fourier ptychography and (b) the proposed method; L1, L2: lens.

**Fig. 2.** Results of ten sets of simulations. The images in first and second row are amplitudes and phases of the original sample; images in third and fourth row are reconstructed amplitudes and phases by the proposed method; images in fifth and sixth row are reconstructed amplitudes and phases by conventional method.

visible-light experiment, the illumination beam had a wavelength of 632.8 nm, and the lenses had the same focal length of 100 mm. The diffuser with phase distributed in the range of $[0–\pi]$ was placed 20 mm downstream of the confocal plane. The detector used in the simulations consists of $512 \times 512$ pixels with a pixel size of 7.4 $\mu m^2$. The diffuser was shifted to $9 \times 9$ grid with a step size of 0.25 mm and the intensity patterns recorded by the detector was calculated by Eq. (2) added into Gaussian noise of zero mean and standard deviation of 1% of the signal value. We run the algorithm described in Section 2 100 cycles and the corresponding reconstructed amplitudes and phases of the sample are shown in 3rd and 4th row of Fig. 2, respectively.

To compare the results of the proposed method and the results of conventional aperture-scanning Fourier ptychography, we performed simulations using the optical setup illustrated in Fig. 1(a). The diameter of the aperture used in simulation had a diameter of 2 mm. The aperture was shift to $9 \times 9$ grid with a step size of 0.25 mm. The recorded images are calculated by Eq. (1) added into Gaussian noise of zero mean and standard deviation of 1% of the signal value. By running the conventional Fourier ptychographic phase retrieval algorithm [37] 100 iterations, the reconstructed amplitudes and phases of the samples for each set of simulation are shown in 5th and 6th row of Fig. 2, respectively. We can find that the reconstructions by the proposed

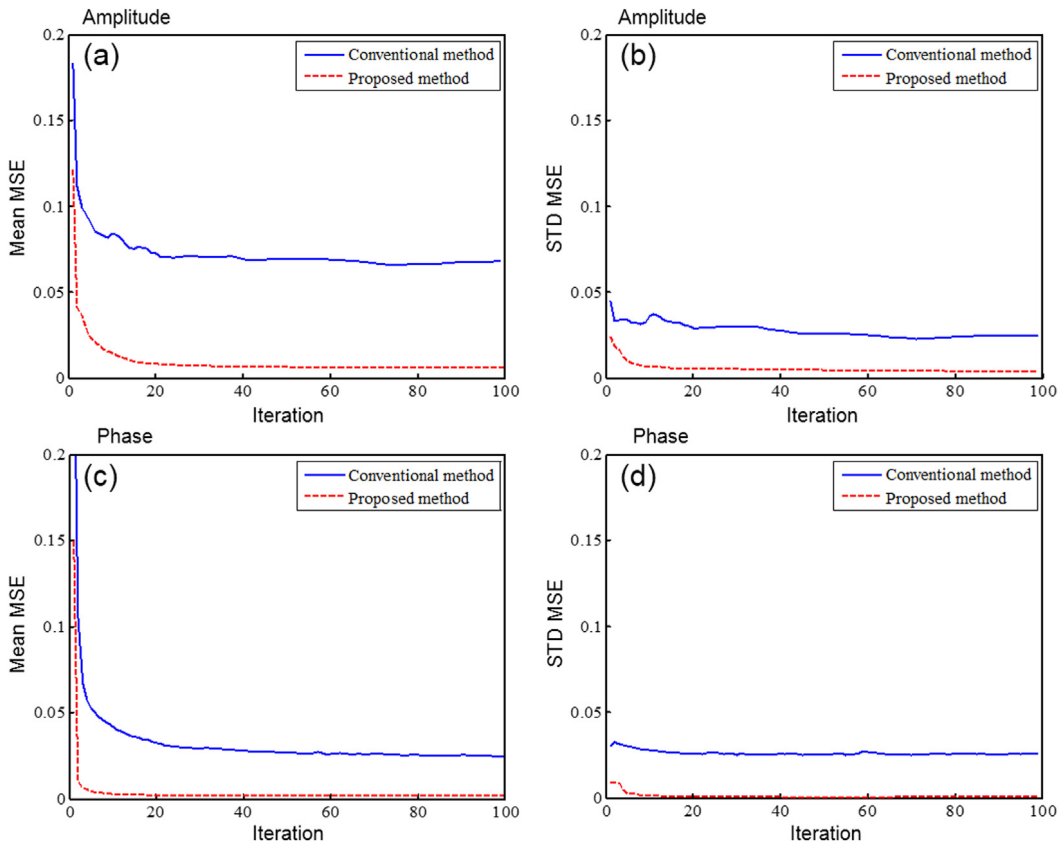


**Fig. 3.** The mean and standard deviation of MSE over ten sets of simulations. (a) Mean and (b) standard deviation of MSE of the reconstructed amplitude; (c) mean and (d) standard deviation of MSE of the reconstructed phase.

method are better than the conventional method when comparing 3rd/4th row images with 5th/6th row images.

To quantitatively evaluate the reconstruction, we calculated the mean squared error (MSE) between the recovered image and the original image. Here we define the MSE as:

$$\mathrm{MSE} = \frac{\sum_1^N (f(x,y) - o(x,y))^2}{N}, \tag{13}$$

where $f(x, y)$ and $o(x, y)$ are the original and reconstructed image of the sample, respectively, $N$ is the number of pixels representing the image. After 100 iterations the calculated MSE values are shown below each reconstructed image in Fig. 2. The MSE values of the images reconstructed by the proposed method are lower than those by conventional aperture-scanning Fourier ptychography, indicating that the proposed method has better reconstructions. The mean and standard deviation of MSE of the reconstructed amplitude over all ten simulations are shown in Fig. 3(a) and (b), respectively. The corresponding MSE of the reconstructed phase over all simulations are shown in Fig. 3(c) and (d). It can be seen from Fig. 3(a) and (c) that the mean MSE of the reconstruction by the proposed method (dashed red line) is lower than the MSE (blue line) by the conventional aperture-scanning method, which means that higher quality reconstructions are obtained by the proposed method. Besides, the standard deviation of MSE by the proposed method [dashed red line in Fig. 3(b) and (d)] is lower than the value by the conventional method [blue line in Fig. 3(b) and (d)], it can be concluded that the proposed method has better convergence stability.

A phase type resolution chart as shown in Fig. 4(a) was used as the original sample to quantitatively compare the resolution of the proposed method with the conventional method. With the same parameters described above, the reconstructed phase by the proposed method is shown in Fig. 4(b1) and the magnified image of red box in Fig. 4(b1) is shown in Fig. 4(b2). The reconstructed phase by conventional method is shown in Fig. 4(c1) and the magnified image of the green box in Fig. 4(c1) is shown in Fig. 4(c2). We can clearly distinguish group 1 in Fig. 4(b2) but it is hardly to distinguish the first group in Fig. 4(c2).

## 4. Experimental results

Optical experiments were carried out to evaluate the effectiveness of the proposed method. We built an experimental imaging system as shown in Fig. 1(b) by using a He-Ne laser with a wavelength of 632.8 nm. The laser beam passed through a spatial filter consists of an objective lens with a magnification of 40× and a pinhole with a diameter of 10 μm, and was then expanded and collimated by a lens with 25 mm in diameter and 100 mm in focal length. A sample under test was placed 100 mm upstream of lens L1, the lenses L1 and L2 that make up the 4f system were the same and had a focal length of 100 mm. A holographic diffuser (1° diffusing angle, Edmund) fixed on a 2D motorized translation stage (Mt1-Z8, controlled by KDC101 DC motor controller, Thorlabs) was placed 20 mm downstream of the confocal plane to modulate the wavefront. In the experiments we placed a 14-Bit CCD detector (Allied Vision Technologies Pike F421B, 2048 × 2048 pixels with pixel size of 7.4 $\mu m^2$) 100 mm downstream of L2 to record the intensity images.

We used the resolution target (USAF 1951) as the sample to test the resolution of the imaging system. The diffuser was shifted to 13 × 13 grid with a step size of 0.25 mm. One of the recorded images is shown in Fig. 5(a). The phase retrieval algorithm described in Section 2 were carried out for image recovering. After 100 iterations, the reconstructed amplitude and phase of the transmission function of the diffuser are shown in Fig. 5(b) and (c). The reconstructed target image is shown in Fig. 5(d). The magnified image of the green box in Fig. 5(d) is shown

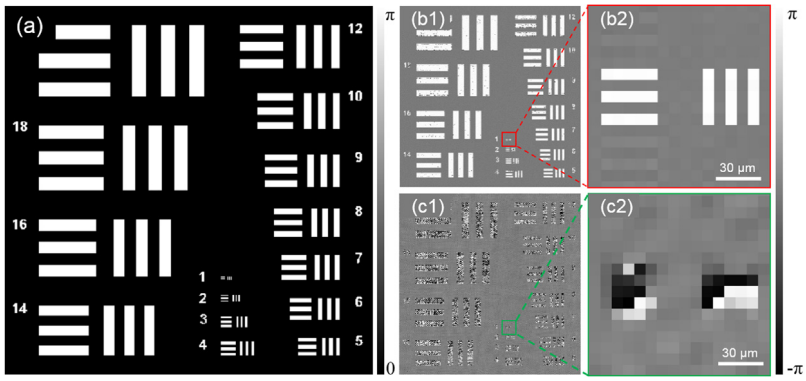


**Fig. 4.** (a) The original resolution target; (b1) reconstructed image by the proposed method, (b2) magnified image of the red box in (b1); (c1) reconstructed image by conventional method, (c2) magnified image of the green box in (c1).

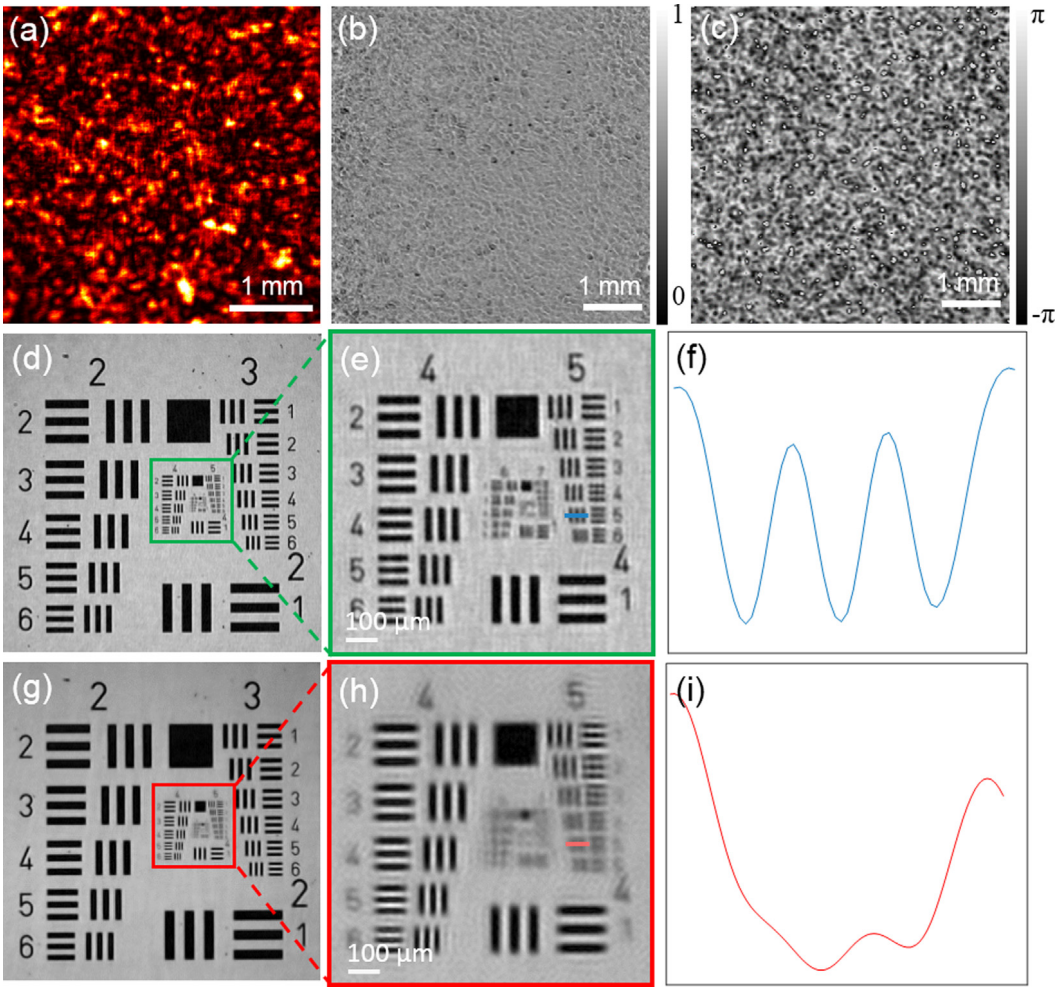


**Fig. 5.** (a)–(f): Results of the proposed method, (a) one of the recorded image, reconstructed (b) amplitude and (c) phase of the diffuser, (d) reconstructed target image, (e) magnified image of the green box in (d), (f) line profile of the blue line in (e); (g) –(i): results of the conventional method, (g) reconstructed target image, (h) magnified image of the red box in (g), (i) line profile of the red line in (h).

in Fig. 5(e), in the image the fifth element in group 5 can be resolved. The line profile of fifth element of group 5 is shown in Fig. 5(f). To compare with the results of conventional method, we built an aperture-scanning experimental setup by scanning an aperture of a diameter of 2 mm in the confocal plane of the 4f system. The aperture was shifted to 13 × 13 positions with the same stepsize as the proposed method. The reconstructed image is shown in Fig. 5(g) and the magnified image of the red box is shown in Fig. 5(h). The smallest detail that can be resolved in Fig. 5(h) is the third element in group 5, thus the resolution is lower than the proposed method.

Another two experiments were conducted using different samples to compare the proposed method with the conventional method. Firstly, a step-height random phase plate was selected as the sample to repeat the above experiment. The reconstructed amplitude and phase of the sample by the proposed method are shown in Fig. 6(a) and (b), while the results by the conventional method are shown in Fig. 6(c) and

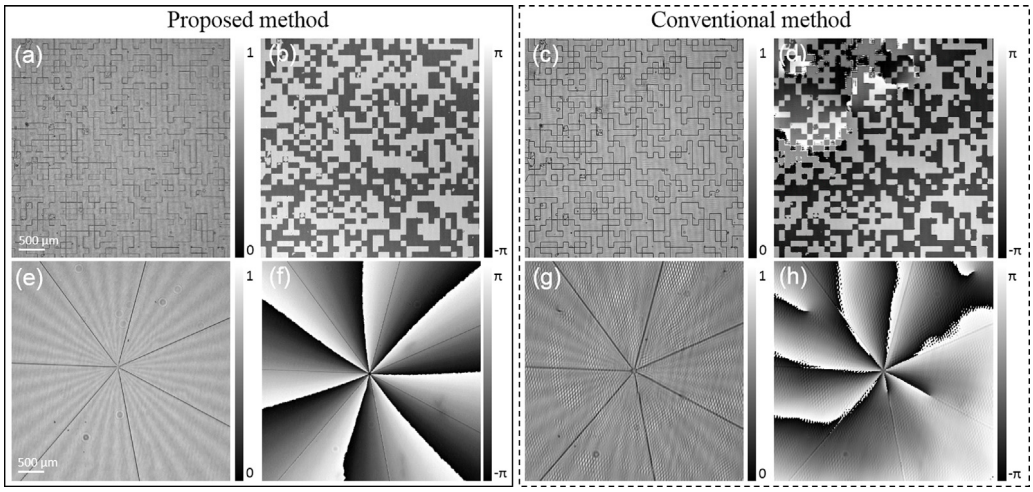


**Fig. 6.** Experiment results using (a)–(d) random phase plate and (e)–(h) vortex phase plate; (a), (b), (e), and (f) results by the proposed method; (c), (d), (g), and (h) results by conventional method.

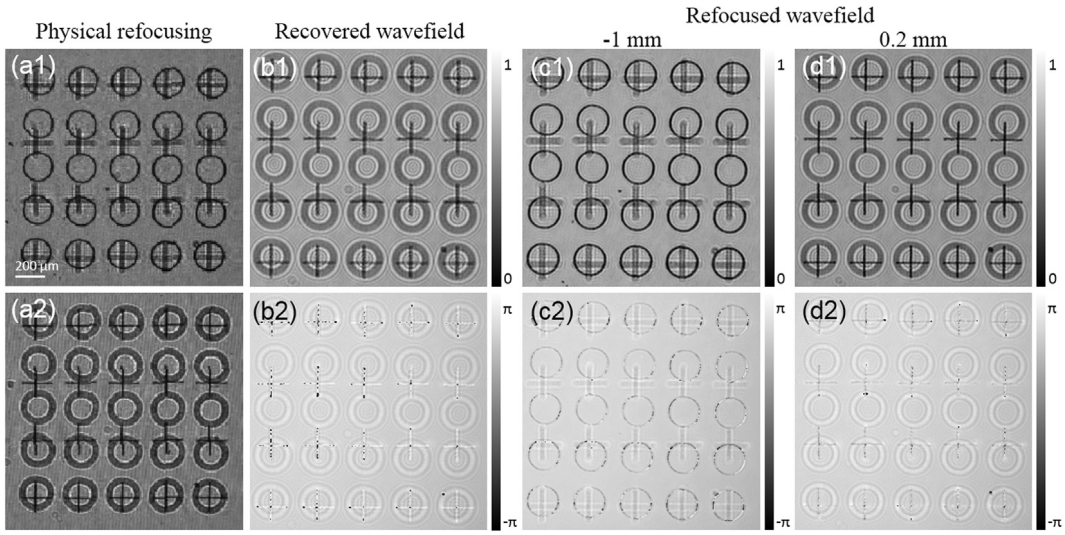


**Fig. 7.** (a1)–(a2) Physical refocusing of each side of a thick glass slide; (b1)–(b2) recovered amplitude and phase of the wavefield of the slide by the proposed method; refocused wavefield by digitally propagating the recovered wavefield to distances of (c1)–(c2) −1 mm and (d1)–(d2) 0.2 mm.

Fig. 6(d), respectively. Secondly, a vortex phase plate (VPP-m633, RPC Photonics) was selected as another sample to evaluate the performance of the proposed method. Vortex phase plates are commonly used optical element for producing vortex beam which has been widely used for applications such as optical tweezers, quantum information processing, optical communication, and high-resolution imaging [45–47]. The recovered amplitude and phase of the vortex phase plate by the proposed method are shown in Figs. 6(e) and 7(f), and the results by conventional method are shown in Fig. 6(g) and (h), respectively. One can observe that the reconstructed phases by the conventional method are distorted when comparing Fig. 6(b)/(f) with Fig. 6(d)/(h).

To demonstrate the feasibility of the proposed method for a thick sample, a 1.2 mm thick glass slide with cross lines and circular lines on each side was imaged using the proposed method. The physical refocused images of each side of the slide are shown in Fig. 7(a1) and (a2), respectively. The recovered amplitude and phase of the wavefield of the sample are shown in Fig. 7(b1) and (b2). The recovered wavefield was digitally propagated to distances of −1 mm and 0.2 mm, allowing different sides of the thick sample to focus, the refocused wavefields are shown in Fig. 7(c1)–(c2) and (d1)–(d2).

## 5. Conclusion

In conclusion, we have proposed and experimentally demonstrated a Fourier ptychographic imaging method via wavefront modulation with an unknown diffuser. By scanning the diffuser, the complex amplitude of the sample and diffuser can be simultaneously reconstructed in this method, and the proposed method has been demonstrated to be more robust to the noise compared to the conventional aperture-scanning Fourier ptychographic method using an aperture. Experimental results show that the presented method is able to measure the phase information of the sample being inspected.

It is worth to point out that in conventional aperture-scanning Fourier ptychography, if a large aperture is used and shifted in the confocal plane, the low frequency component of the sample will always passes through the imaging system, and no dark images would be recorded, thus the recorded images have high SNR. However, in this case, the recorded images lack data diversity, which will result in slow convergence and low reconstruction accuracy. In our method, the low frequency and high frequency components of the sample spectrum are both modulated at each position of the diffuser, which results in high data diversity at the detector plane. Therefore, the presented method has better convergence and high reconstruction accuracy.

In our experimental implement of the method, a holographic diffuser is mechanically scanned which introduces location errors of the diffuser and increases the data acquisition time, an alternate implement of the proposed method is to replace the holographic diffuser with a spatial light modulator (SLM) or a digital micromirror device (DMD) to create shifted random structures. Our future work in this field will investigate how to further accelerate the convergence by adding Nesterov's momentum [44] and realize depth resolved imaging with the combination of compressive sensing techniques [36].

The method will be used for applications such as wavefront sensing, wafer inspection, and optical testing.

## CRediT authorship contribution statement

**Xiaoliang He:** Conceptualization, Writing - original draft, Methodology, Validation. **Zhilong Jiang:** Methodology, Validation. **Yan Kong:** Validation. **Shouyu Wang:** Validation. **Cheng Liu:** Conceptualization, Supervision.

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